

Chih-Wei Joshua Liu

+1 (617) 580-7953 · cjoshliu@stanford.edu · GitHub: [cjoshliu](#)

EDUCATION

Stanford University Biophysics PhD Program · Thesis advisor: Grant M. Rotskoff	Stanford, CA, USA 09/2024 – present
Massachusetts Institute of Technology (MIT) Bachelor of Science · Major: Physics · GPA: 4.9/5.0 · MCAT: 528/528	Cambridge, MA, USA 05/2022
National Experimental High School SAT: 1600/1600 · ACT: 36/36	Hsinchu, Taiwan, ROC 05/2018

EXPERIENCE

MIT Physics of Living Systems Group	Research Assistant (06/2022 – 08/2024)
MIT Department of Physics	Teaching Assistant, 8.01 Physics I (09/2019 – 12/2019)
	Teaching Assistant, 8.241 Introduction to Biological Physics (09/2023 – 05/2024)
Massachusetts General Hospital	Data Analysis Intern (06/2021 – 08/2021)

Analyzed patient statistics and authored two clinical publications.

SELECTED PROJECTS

C.-W.J. Liu, J. Klinger, G.M. Rotskoff, Optimal parameterization of nonequilibrium generalized master equations from discrete-time experimental data, [arXiv:2606.28289](#), in revision at *Phys. Rev. X* (2026). [\[code\]](#)

- Derived the first maximum-likelihood parameterization of non-Markovian kinetics satisfying all physical constraints in general biomolecules. Validated on experimental recordings of CFTR ion channels and F1 molecular motors.
- Set new state-of-the-art in benchmark extraction of HP35-Lys24Nle/Asn27His/Lys29Nle protein-folding kinetics from coarse-grained models trained on atomistic molecular-dynamics simulation.

J. Li*, **C.-W.J. Liu***, M. Szurek, and N. Fakhri, **Measuring irreversibility from learned representations of biological patterns**, *PRX Life* **2**, 033013 (2024), *equal contribution. [\[code\]](#)

- Quantified energetic costs of actomyosin regulation by Rho GTPase using only variational-autoencoder representations of cortical-wave microscopy videos.

A. Yona, **C.-W.J. Liu**, Z. Ben-Moshe, T.M. Nichols, G.A. Bodhankar, F.M. Key, C.V. Rao, T. Lieberman, D. Tullman-Ercek, and J. Gore, **Adaptive evolution identifies candidate genetic routes associated with oxalate utilization in probiotic and gut-derived *E. coli***, in revision at *Cell Rep. Methods* (2026).

- Led biochemical characterization of human gut microbes subject to directed evolution for nephrotoxin degradation, identifying loci repeatedly mutated across lineages through whole-genome sequencing and imaging assays.

H. Lee*, E. Koskelo*, S. Gokhale*, J. Li*, C. Fei, **C.-W.J. Liu**, L. Lin, J. Dunkel, D.J. Skinner, and N. Fakhri, **Topological flowscape reveals state transitions in nonreciprocal living matter**, *Phys. Rev. X* **16**, 031061 (2026), *equal contribution.

- Trained classifier of developmental stages in embryo videos, enabling identification of novel swarming dynamics in multicellular aggregates. Featured in *Physics* magazine and on the cover of *Phys. Rev. X*.

LEADERSHIP AND HONORS

National Science Foundation Graduate Research Fellowship (GRFP, 2024, Physics of Living Systems)
National Defense Science and Engineering Graduate Fellowship (2024, Physics, declined for GRFP)
Joel Matthew Orloff Award (MIT Department of Physics, 2022) · **MIT News:** [profile](#) (2022)
USA Biology Olympiad: High Honorist (2017); Semifinalist (2015–2018)
MIT Society of Physics Students and Sigma Pi Sigma: honor-society inductee (2022); President (2022)
Phi Kappa Theta – Massachusetts Eta: Pledgemaster (2021); sourced and fundraised donation of 12,000 surgical masks to Beth Israel Deaconess Medical Center during pandemic shortage as Service Chair (2019–2020).

SKILLS AND INTERESTS

Computing: lightning, jax, pydantic, numpy, scipy, torch, scikit-learn, statsmodels, uv, ty, ruff, modal
Laboratory: cell culture, molecular cloning, light microscopy, NMR spectroscopy, anaerobic technique
Research interests: nonequilibrium thermodynamics, biochemical kinetics, molecular motors, optimal transport
Personal interests: phonology, historical linguistics, ichthyology · **Natural language:** Standard Chinese (native)